

AWP Broiler Forecast App

User Guide

Al-Watania Poultry — Production Planning & S&OP

1. Introduction

The **AWP Broiler Forecast App** is a web-based Sales and Operations Planning (S&OP) tool built exclusively for AI-Watania Poultry (AWP). It enables production planners and supply chain managers to translate weekly market demand into a day-by-day chick placement schedule, trace every bird through the processing chain, and produce a fully reconciled S&OP plan in minutes.

The app covers the complete planning cycle: Demand entry → Supply calculation → Reconciliation → Demand-Driven Placement → S&OP Report, alongside a six-step Production Pipeline that models the broiler cycle from chick placement to slaughtered carcass weight.

Who is this guide for? Production planners, S&OP coordinators, and supply chain managers at AWP. No technical background is required.

1.1 Key Benefits

- Eliminates manual spreadsheet calculations — the entire processing chain recalculates instantly.
- Demand-first architecture: start with what the market needs, not what farms can produce.
- One-click Excel export for every planning step — ready for SAP upload or management reporting.
- Scenario comparison: save up to 3 named plans and compare them side by side.
- Configurable parameters — every AWP operational constant is editable without touching code.

2. Getting Started

2.1 Accessing the App

Open a modern desktop or tablet browser (Chrome, Edge, or Firefox recommended) and navigate to the app URL provided by your IT team or the Vercel deployment link. **A minimum screen width of 768 px is required** — the app is optimised for 1440 px wide displays. Mobile phones show a read-only summary view.

2.2 App Layout

The app has three persistent regions:

Region	Purpose
Left Sidebar	Navigation between S&OP; Modules (M1–M5), Production Pipeline Steps (1–6), and Scenario Comparison. Click the circular  tab on the sidebar edge to collapse it and gain screen space.
Top Bar	Shows the current view name, an Assumptions button (opens the Parameter Panel), and Export buttons (Excel / PDF).
Main Content Area	Displays the currently selected module or pipeline step. Scrolls independently.

2.3 Parameter Panel (Assumptions)

Click **Assumptions** in the top bar to open the Parameter Panel on the right side. This panel contains every configurable AWP operational constant: cycle length, mortality rates, weight parameters, plant shares, and more. Changes here instantly cascade to all downstream calculations. Click **Reset to Defaults** to restore the factory AWP values.

3. Screen-by-Screen Guide

3.1 Home Dashboard

The Home Dashboard is the landing screen. It displays a hero banner followed by:

- **KPI Strip:** 6 summary cards — Chicks Placed, Harvestable Birds, Total Carcass (kg), Total Production (kg), Avg Utilization %, and Weeks Over Capacity.
- **S&OP; Module Cards (M1–M5):** Each card shows live metrics for that module and an Open → button to navigate directly to it.
- **Production Pipeline Cards (Steps 1–6):** Each card previews the key output of that pipeline step with an embedded mini-chart or data snippet.

All numbers on the dashboard update in real time as you change demand, parameters, or the placement plan.

3.2 M1 — Demand Plan

The Demand Plan is a product × sales-channel × week matrix. It is the starting point of the S&OP; cycle.

What you see

- 8 channel tabs across the top: Distributors, Export, Food Servers, Modern Trade, Sister Companies, Traditional Trade, Wholesale, E-commerce. Plus an **All** summary tab.
- Products listed as rows, grouped by category: Whole Chicken (Grade A/B, weight buckets, fresh/frozen), Cuts, FPP, and Eggs.
- Week columns (up to 26 weeks) — each cell is editable in-place.
- A row-total column and summary cards above the matrix showing total tons per category.

How to enter demand

- Select a channel tab.
- Click any cell in the matrix and type the quantity in **metric tons** (eggs in trays).
- Press Tab to move to the next week or Enter to confirm.
- Use the % **Adjust** toolbar button to apply a percentage change to a product/channel/week range at once.
- Use **Copy Week Forward** to repeat a week's values across the rest of the horizon.

Import from SAP

Click the **Import Sales Plan** button to open the SAP Import Panel. Upload your SAP-exported Excel or CSV file. The app auto-maps SAP row signatures to products and SAP channel names to the 8 AWP channels. Review the mapping table, adjust any mismatches, then click **Apply to Demand Plan**.

3.3 M2 — Supply Requirements

This module reverse-engineers the demand plan: given weekly demand in tons, it calculates how many chicks must be placed to satisfy each category.

What you see

- A yield information banner showing the WC, FPP, and Cuts yield fractions (derived from your current family allocation and cut yields).
- A weekly table with columns: WC Demand (t), FPP Demand (t), Cuts Demand (t), Required Carcass (kg), Required Harvestable Birds, Required Chicks to Place, Planned Carcass (kg), and a **Gap pill** — green (Surplus), amber (Tight ±5%), or red (Deficit).
- The **Binding Category** column shows which product family is driving the requirement in each week.

3.4 M3 — Reconciliation

The Reconciliation dashboard compares the demand plan against the supply plan and highlights gaps by category.

What you see

- Three summary cards at the top — one each for Whole Chicken, FPP, and Cuts — showing total demand, total supply, and the net gap in kg.
- A stacked bar chart showing supply by category per week with a demand overlay line.
- A full week-by-week gap table with status pills: Surplus / Tight / Deficit.
- Deficit and surplus week counts in the summary header.

3.5 M4 — Demand-Driven Placement

Module 4 translates the supply requirement into a concrete placement schedule. Instead of manually filling the placement grid, the app calculates exactly how many houses to place per day to meet the demand plan.

How it works

- The required chicks-to-place per harvest week (from M2) is divided by the working days in that week (excluding Fridays if Friday Off is enabled).
- The harvest week is shifted back by the cycle length ($\lceil 25.5 \div 7 \rceil = 4$ weeks) to find the correct **placement week**.
- The preview table shows **Current Houses/Day** vs **Required Houses/Day** per week, with over-capacity and pre-plan warnings highlighted.
- Click **Apply to Placement Plan** to write the required values directly into the Step 1 placement calendar.

After clicking Apply, navigate to Step 1 (Placement Plan) to review and fine-tune the day-by-day values before running the full pipeline.

3.6 M5 — S&OP; Report

The S&OP; Report is the executive summary of the planning cycle. It consolidates demand, supply, and reconciliation into a single RAG (Red / Amber / Green) dashboard.

What you see

- **Traffic-light summary cards** — one per product category showing overall RAG status.
- **KPI strip** — count of Deficit weeks, Tight weeks, Pre-Plan weeks, and OK weeks.
- **Full S&OP; table** — week by week with per-category RAG dots and an Overall Status pill.
- **Exceptions list** — auto-generated action items for every week in Deficit or Tight status.

Exporting the S&OP; Report

Use the **Export PDF** button to download a print-ready PDF, or **Print** to send directly to a printer or save as PDF via the browser print dialog.

4. Production Pipeline — Step-by-Step

The Production Pipeline simulates the physical broiler cycle. Each step feeds the next. All steps are read-only outputs except Step 1 (placement input) and the inline parameter fields in Steps 3–5.

Step 1 — Placement Plan

The Placement Plan is a day-by-day calendar for the planning horizon. Each row represents one calendar day. You enter the number of **farms placing** and the **chicks per house** for that day.

- Use **Quick Fill** to instantly populate all eligible days with the default house count (configurable in Assumptions).

- Use **Import CSV / Excel** to upload a pre-prepared placement schedule. Download the template first to see the required format.
- Friday rows are automatically greyed out when Friday Off is enabled.
- Adjust the planning horizon (8–26 weeks) using the selector at the top.

Step 2 — Live Bird Forecast

Cascades from the placement plan. Shows how many harvestable birds will be available each week, total live weight, and plant utilization.

- **Capacity Chart:** weekly harvestable birds (bars) vs total plant capacity (line). Red bars indicate weeks where capacity is exceeded.
- **Processing Funnel table:** 8-stage waterfall from Harvestable → Slaughtered Carcass, with percentage of harvestable birds for each stage.
- **Funnel chart:** visual representation of the 4 main stages.

Step 3 — Carcass Yield & Grade Split

Distributes slaughtered carcass weight across Grade A, B, and C, and across 11 weight classes (500g to 1500g).

- Grade split percentages are editable inline (A / B / C must sum to 100%).
- Carcass size distribution percentages are also editable inline; the sum indicator turns red if they do not total 100%.
- **Grade Chart:** stacked bar showing kg per grade per week.

Step 4 — Product Family Allocation

Allocates carcass kg per grade into three product families: Whole Chicken Fresh, Whole Chicken Frozen, and FPP.

- The 3×3 allocation matrix (Grade A/B/C × WC-Fresh/WC-Frozen/FPP) is fully editable.
- Each grade row must sum to 100% or the validation banner will warn you.
- **Family Donut chart:** shows the WC Fresh / WC Frozen / FPP split over the horizon.

Step 5 — FPP Cut Plan

Breaks the FPP kg per week into individual cuts using configurable yield percentages.

- Toggle **Leg Split Mode** to switch between Whole Leg and Drumstick + Thigh Bone-In.
- Cut yield percentages are editable inline; the total indicator shows how much of FPP input is accounted for (target: close to 100%).
- The weekly table shows kg per cut per week. The top 3 cuts by total volume are highlighted.

Step 6 — Processing Plan by Plant

Distributes birds and carcass across the three slaughter plants based on configured plant share percentages.

- Use the plant tabs (All / Plant 1 / Plant 2 / Plant 3) to filter the table.
- Red rows indicate weeks where daily birds exceed a plant's capacity.
- Average utilisation per plant is shown in the summary card.

4.7 Scenario Comparison

Access via the Scenario Comparison item in the sidebar Tools section. Save up to 3 named snapshots of the current plan (parameters + placement schedule). The comparison table shows 8 key metrics side by side: total chicks placed, harvestable birds, carcass kg, FPP kg, WC fresh kg, WC frozen kg, average utilisation %, and weeks over capacity. Use this to evaluate the impact of different planning assumptions before committing.

5. Calculation Logic

All calculations execute in the browser in real time. The pipeline runs automatically whenever any input changes. There is no Save button — everything is live.

5.1 Processing Chain

#	Output	Formula	Default Rate / Value
1	Harvestable Birds	Chicks Placed × (1 – Mortality %)	5.00%
2	Harvest Mortality Birds	Harvestable Birds × 0.20%	0.20%
3	Dispatched Birds	Harvestable Birds – Harvest Mortality	—
4	Dead on Arrival (DOA)	Dispatched Birds × 0.50%	0.50%
5	Culled Birds	Dispatched Birds × 0.20%	0.20%
6	Electronic Bird Count	Dispatched – DOA – Culled	—
7	Plucking Rejects	Electronic Bird Count × 0.60%	0.60%
8	Slaughtered Birds	Electronic Bird Count – Plucking Rejects	—
9	Total Live Weight (kg)	Harvestable Birds × ALW (1.220 kg)	ALW = 1.220 kg
10	Slaughtered Carcass (kg)	Slaughtered Birds × ACW (0.830 kg)	ACW = 0.830 kg
11	Carcass Yield %	ACW ÷ ALW	68.03%

6. Operational Parameters

All default values reflect AWP's confirmed operational parameters. They are editable in the Assumptions panel and reset via the Reset to Defaults button.

Parameter	Default Value	Description
Chicks per House	28,500	Average placed birds per house
House Count per Farm	27	Houses per farm
Broiler Cycle Length	25.5 days	Placement to harvest
Mortality Rate	5.00%	Farm mortality during grow-out
Harvest Mortality Rate	0.20%	Birds lost at catching
Dead on Arrival Rate	0.50%	Birds dead in transit
Culled Bird Rate	0.20%	Birds culled at line entry
Plucking Reject Rate	0.60%	Birds rejected at plucking
Avg Live Weight (ALW)	1.220 kg	Live weight at harvest
Avg Carcass Weight (ACW)	0.830 kg	Carcass weight post-processing
Carcass Yield %	68.03%	ACW ÷ ALW (derived, not input)
Plant 1 Share	15%	Fraction of birds to Plant 1
Plant 2 Share	25%	Fraction of birds to Plant 2
Plant 3 Share	60%	Fraction of birds to Plant 3

Parameter	Default Value	Description
Planning Horizon	16 weeks	Default horizon (8–26 weeks)
Working Days per Week	6	Saturday–Thursday (Friday Off)
Friday Off	Enabled	Placement and catching suspended on Fridays

6.1 Carcass Size Distribution (Default)

Weight Class	% of Birds	Weight Class	% of Birds
500g	1.35%	600g	5.00%
700g	16.30%	800g	33.00%
900g	30.50%	1000g	11.60%
1100g	2.00%	1200g	0.20%
1300g	0.02%	1400g	0.02%
1500g	0.01%		

7. Exporting Data

Click the **Export Excel** button in the top bar to download a fully formatted workbook. The workbook contains one sheet per pipeline step:

Sheet	Contents
Placement Plan	Day-by-day farm placing and chick counts
Live Bird Forecast	Weekly harvestable birds, live weight, utilisation
Processing Funnel	8-stage funnel: harvestable → slaughtered carcass
Carcass Yield	Weekly carcass kg and grade A/B/C split
Size Distribution	Weekly birds per 11 weight classes
Product Family	Weekly WC Fresh, WC Frozen, FPP kg
FPP Cut Plan	Weekly kg per cut type
Processing by Plant	Weekly birds, carcass, family kg per plant

The **Demand Plan export** (from M1) produces a separate workbook with one sheet per sales channel plus an All-channels summary sheet.

The **S&OP; Report export** (from M5) produces a workbook with the weekly S&OP; table and a summary tab.

Download the Placement Template or SAP Sales Plan Template buttons in their respective screens to get a correctly formatted import file.

8. Glossary

Term	Definition
ACW	Average Carcass Weight — weight of the processed carcass (default 0.830 kg)

Term	Definition
ALW	Average Live Weight — live bird weight at harvest (default 1.220 kg)
DDP	Demand-Driven Placement — Module 4; back-calculates chick placement from demand
DOA	Dead on Arrival — birds that die in transit from farm to plant
EBC	Electronic Bird Count — automated line count after DOA and culled removal
FPP	Further Processed Products — deboned/portioned products (cuts, nuggets, etc.)
LSMW	Legacy System Migration Workbench — SAP tool for mass data upload
MAPE	Mean Absolute Percentage Error — forecast accuracy metric
MEQ1	SAP transaction for maintaining quota arrangements in MRP
MRP	Material Requirements Planning — SAP planning engine
RAG	Red / Amber / Green — traffic-light status system for S&OP; exceptions
S&OP;	Sales and Operations Planning — integrated demand and supply balancing process
SAP	Systems, Applications & Products — ERP system used by AWP
WC	Whole Chicken — carcass sold fresh or frozen without further processing

9. FAQ & Troubleshooting

Q: Can I use the app on my phone?

A: The app is designed for desktop and tablet screens (768 px or wider). On smaller screens it shows a read-only summary. For full planning functionality use a laptop or desktop.

Q: My numbers don't match the SAP report — what should I check?

A: Verify that the Assumptions panel shows the correct AWP parameters (cycle length, mortality rates, ALW, ACW). Then check that the Planning Horizon and Plan Start Date are correct. Also confirm your demand plan channels map correctly to SAP channels in the Import Panel.

Q: How do I reset everything to factory defaults?

A: Open the Assumptions panel (top bar → Assumptions) and click Reset to Defaults at the bottom. This restores all parameters, the placement plan, and the demand plan to their shipped values.

Q: The red validation banner says 'grade split does not sum to 100%' — what do I do?

A: Go to Step 3 (Carcass Yield) and adjust the Grade A / B / C percentage inputs until they sum to exactly 100%. The same applies to the family allocation matrix in Step 4 and cut yields in Step 5.

Q: Why are some placement days greyed out?

A: Friday Off is enabled by default. Friday rows are locked to zero placements. To disable this, open Assumptions and toggle Friday Off to off.

Q: The SAP import panel shows 'No rows matched' — why?

A: The column headers in your SAP export file may differ from the expected format. Download the SAP Sales Plan Template from the Import Panel to see the expected column layout, and ensure your file contains a 'Channels' column with matching channel names.

Q: Can I save multiple planning scenarios?

A: Yes. Navigate to Scenario Comparison in the sidebar, enter a scenario name, and click Save. Up to 3 snapshots can be stored. The comparison table shows key metrics side by side.

Q: Is data saved if I close the browser?

A: Yes. The app uses browser local storage (persist) to save your plan automatically. Data persists across browser sessions on the same device. Clearing browser data will reset the app.

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